



Preparation and antifouling performance of tin-free self-polishing antifouling coatings based on side-chain suspended indole derivative structural resins

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ABSTRACT

Tin-free self-polishing antifouling coatings have the highest market share since organotin self-polishing antifouling coatings have been banned. However, its high dependence on cuprous oxide was found to have caused potential harm to the environment, making it necessary to improve the functionality of the resin. In this paper, a zinc acrylate resin with side chain hanging indole derivative structure was prepared by using N-(1H-5-bromoindole-3-methylene) (BIAM) with good biological activity as functional monomer. The functional resin with good antifouling performance was selected by antibacterial and algae inhibition experiments. The results showed that when the BIAM content was 9 %, the inhibition rates of the resin on *E. coli* and *Prymnesium parvum* reached 98 % and 90 %, respectively. Tin-free self-polishing antifouling coatings were prepared using the above resins as film-forming materials. The anti-protein adsorption performance and antifouling performance of the coating were tested by anti-protein adsorption experiment and real sea hanging plate experiment. The results showed that the coating containing indole derivative structure had good anti-protein adsorption performance and antifouling performance, and the higher the BIAM content, the better the anti-protein adsorption performance and marine antifouling performance.

1. Introduction

Facilities immersed in the marine environment are susceptible to fouling by marine organisms, which can adversely affect shipping or other marine activities (Liu et al., 2023; Snowdon et al., 2023; Sun et al., 2022a). For example, the attachment of marine fouling organisms can increase the sailing resistance of the ship, leading to an increase in fuel consumption, promote corrosion of the substrate and consequent damage to the hull, clog cooling pipes and reduce their service life, and affect the aquaculture industry and increase farming costs (Reyes et al., 2020; Sarkar et al., 2022; Sievers et al., 2019; Tian et al., 2021a). Among many antifouling technologies, applying antifouling coatings is the most economical and effective way to solve the problem of marine biofouling (Dong et al., 2022).

Traditional antifouling coatings mostly rely on the biological killing effect of antifouling agents such as TBT and DDT to prevent biofouling (Beyer et al., 2022a; Howell and Behrends, 2006; Maia et al., 2015). But this antifouling method has been prohibited due to the significant harm

to marine organisms (Beyer et al., 2022b; Oguri et al., 2017; Ytreberg et al., 2016). Therefore, a new type of antifouling coating has emerged to realize the excellent antifouling performance of the coating with the help of the functionality of the resin. Among the new antifouling coatings, tin-free self-polishing antifouling coatings drawing on organotin antifouling mechanism occupy a major share of the market due to their advantages of high antifouling activity, long antifouling period, low price, etc. However, most of the tin-free self-polishing antifouling coatings need to add a large amount of Cu₂O as antifouling agent (Lavtizar et al., 2018; Tian et al., 2021b). Studies have shown that Cu₂O will accumulate in marine organisms, induce harmful mutations in biological genes, endanger the marine environment, and face the problem of being banned in the future (Liu et al., 2017a; Whitacre, 2015). Therefore, improving the antifouling performance of the functional resin itself is the key to the development of new environmentally friendly antifouling coatings.

To further improve the antifouling properties of resins, modification of self-polishing resins is the main way (Wang et al., 2022). Modified

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self-polishing resins are divided into three main categories: self-polishing resins containing antimicrobial agents (Sha et al., 2023; Sun et al., 2022b), biodegradable self-polishing resins (Ali et al., 2021; Fay et al., 2007; Zhou et al., 2015), and low surface energy self-polishing resins (Martinelli et al., 2016; Sun et al., 2015; Yesudass Sebastin et al., 2019). Among them, self-polishing resins containing antibacterial agents have been extensively studied. The side chains of these resins contain functional groups with biocidal activity, which can be released into the marine environment through hydrolysis to achieve high antifouling activity of the resins themselves (Omae, 2003).

Dong et al. introduced heterocyclic compounds and benzisothiazolinones into acrylic resins, and the results of performance tests showed that the introduction of new functional structure significantly improved the antibacterial and marine antifouling properties of the resins (Dong et al., 2021). Chen et al. prepared main-chain benzoxazine (MCBZ) resins, and the antimicrobial, algal inhibition and real-sea antifouling experiments proved that MCBZ resins have excellent antifouling properties (Chen et al., 2023). Nasreena et al. introduced urushiol into PMMA to prepare an antifouling coating, and it was demonstrated by bacteriostatic experiments that lac phenol could significantly improve the bacteriostatic properties of the coating (Lone et al., 2017). Guo et al. prepared a polyurethane with 7-amino-4-methylcoumarin as the antifouling group, and the performance test results proved that the polyurethane had good bacterial inhibition and excellent marine antifouling properties (Guo et al., 2021). Sun et al. introduced active substances extracted from *Zanthoxylum bungeanum* into the resin and demonstrated that the resin has good marine antifouling properties (Sun et al., 2022c).

At present, the research on the modified self-polishing resin is extensive, but there are few reports on the research of this kind of resin in antifouling coatings, so these studies have not played a guiding role in the further development of new tin-free self-polishing antifouling coatings. In this study, considering the excellent antifouling properties of indole natural products (Chen et al., 2018; He et al., 2013; Kon-ya, 1994; Olguin-Urbe et al., 1997) and their analogues (Feng et al., 2019a; Li et al., 2014; Ni et al., 2021; Yang et al., 2014), N-(1H-5-bromoindole-3-ylmethylene) (BIAM) (Li et al., 2009), which has been verified to have good biological attachment inhibition activity, was used as a functional monomer to prepare zinc acrylate resins with side-chain-suspended indole derivatives. The low toxicity and long-lasting self-polishing antifouling coatings were prepared by using the above resin as film-forming materials, and the application performance of the antifouling coatings was studied, which provided theoretical and technical basis for the development of new self-polishing antifouling coatings with low cuprous oxide content.

2. Experimental

2.1. Materials

N-(1H-5-bromoindole-3-methylene) (BIAM) acrylamide was synthesized according to the literature (Li et al., 2009). Ethyl acrylate, butyl acrylate, acrylic acid, anhydrous ethanol, butyl acetate, Anhydrous aluminum chloride and azobisisobutyronitrile (AIBN), Cu₂O and ZnO were purchased from Shanghai Aladdin Biochemical Technology Co. Phosphate buffer solution (PBS) and bovine serum albumin (BSA) were purchased from Shanghai Macklin Biochemical Technology Co.

2.2. Synthesis of zinc acrylic resins containing indole derivative structure

First, the solvent (40 % of prepolymer, wt%) xylene was mixed well with n-butanol in a mass ratio of 4:1 and added to a three-necked flask with an electric stirring device.

Then, acrylic monomers, N-(1H-5-bromoindole-3-methylene) and AIBN (1.2 % of prepolymer, wt%) were mixed evenly and added every 15 min. The temperature of the reaction system was 85 °C, and the

prepolymer was obtained by holding for 4 h. The prepolymer is a polymer obtained by copolymerization of acrylic monomers and N-(1H-5-bromoindole-3-methylene).

Finally, ZnO, benzoic acid/ naphthenic acid and solvent (n-butanol: xylene: butyl acetate = 1:4.4:10.6, mass ratio) were added to the prepolymer. The reaction was carried out at 85 °C for 4 h and then dehydrated at 110 °C. The reaction system became transparent. Zinc acrylate resins containing indole derivative structures (R-B-x/R-N-x, x = 0, 3, 6, 9) (R-B/N refers to resins with benzoic acid/naphthenic acid side chains, respectively) were then obtained and characterized by IR. The synthetic route for the zinc acrylate resins containing indole derivative structures is shown in Fig. 1.

The compositions of the resins are shown in Table 1. The molar fractions of N-(1H-5-bromoindol-3-ylidene) introduced into the zinc acrylate resins were 0 %, 3 %, 6 % and 9 %, respectively.

2.3. Preparation of antifouling paints

As shown in Fig. 2, the resin, pigment and filler, solvent and antifouling agent were firstly dispersed homogeneously at a speed of 3000 r/min, and then glass beads were added for grinding. The antifouling coating was prepared after the fineness was ≤80 μm. The basic formula of the antifouling coating is shown in Table 2, the specific formula is shown in Table S1 of the Supplementary materials, and the prepared antifouling coatings are shown in Table 3.

2.4. Antibacterial activity evaluation of zinc acrylic resins

In this work, two types of bacteria, *Escherichia coli* (*E. coli*) and *Pseudomonas tetradonis* (*P. tetradonis*), were selected as model microorganisms, and the colony formation counting method (Feng et al., 2019b) was used to evaluate the antibacterial performance of acrylic resin. Prior to the antimicrobial experiments, all glassware was sterilized in an autoclave and all acrylic resin films (50 mm × 50 mm) were sterilized under UV light for 30 min. *P. tetradonis* or *E. coli* on fresh agar plates were incubated in 2216 E liquid medium or nutrient broth for 24 h at 37 °C, and then PBS was diluted with filtered and then sterilized seawater to 10⁸ CFU/mL (as determined by a hemocytometer) by using the standard serial dilution method. 0.2 mL droplet of the bacterial suspension was placed in a zinc acrylic resin film and incubated for 24 h at 37 °C understanding conditions. The bacterial suspension in the resin film was then diluted with PBS to the appropriate concentration. Pipette 200 μL of bacterial suspension on nutrient agar plates with a pipette gun, spread evenly using a triangular blotting rod, and incubate in a constant temperature incubator at 37 °C for 24 h. Calculate the number of surviving colonies on the petri dishes by the tiling colony counting method (Liu et al., 2017b; Shi et al., 2016).

The antibacterial rate was obtained from the following equation:

$$A = \frac{N_0 - N_i}{N_0} \times 100\%$$

where A is the bacteriostatic rate and N₀ and N_i are the number of colonies on the control and experimental agar plates, respectively.

2.5. Algal inhibition test of zinc acrylic resins

The precultured algal (*Gymnodinium aerucyinosu*, *Phaeodactylum tri-cornutum* Bohli, and *Prymnesium parvum*) used as the tested algal species. First, the filtered seawater was sterilized in an autoclave, and the cooled seawater was added with nutrient salts according to Table 4. The test algae were inoculated into a certain amount of nutrient solution and then cultured in an artificial climate incubator. The maximum absorption wavelengths of the three microalgae were obtained by scanning with a UV-visible spectrophotometer. The algae solution was diluted to various levels during the exponential phase of growth, and absorbance

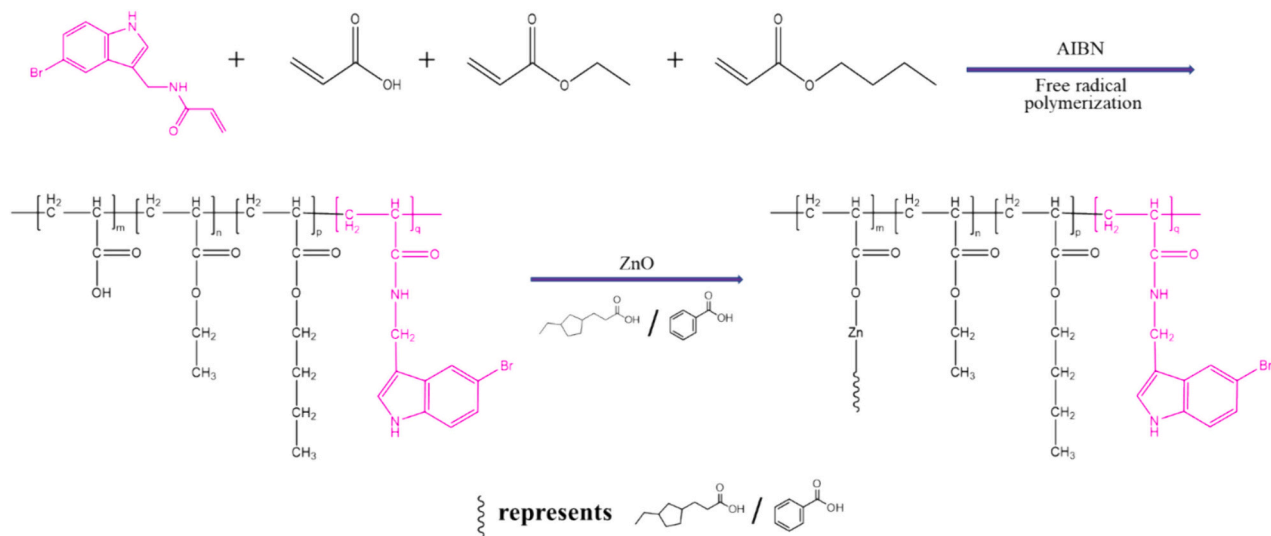


Fig. 1. Synthetic routes of zinc acrylic resins suspending the indole derivative structure.

Table 1
Composition and relative content of resins.

Sample	Monomer ^a					
	BA	EA	AA	Indole derivative	ZnO	Benzoic acid/ naphthenic acid
R-B/N-0	3	3	2	0 (0 %)	2	2
R-B/N-3	3	3	2	0.24(3 %)	2	2
R-B/N-6	3	3	2	0.48(6 %)	2	2
R-B/N-9	3	3	2	0.96(9 %)	2	2

^a The proportion of monomers is molar ratio.

values of algae solution were tested under maximum absorption wavelengths of different algae. Next, the standard working curve was made with the concentrations of algal solution as the ordinate and the absorbance value as the abscissa. In addition, the concentrations of the corresponding algal solution were calculated by a blood cell counting chamber. The resin samples to be tested were sterilized by UV light for 24 h and then placed in a surface dish with diluted algal solution. Pure glass slides were used as blank control and absorbance was measured at the maximum absorption wavelength of the algal solution every day

during seven days of incubation. Algal inhibition was calculated by the following formula (Feng et al., 2019c):

$S = (S_0 - S_i) / S_0 \times 100 \%$, where S is the algal inhibition rate, S_0 is the absorbance value of the control group and S_i is the absorbance value of the experimental group (Ni et al., 2020).

2.6. Self-polishing performance analysis of antifouling paints

As shown in Fig. 3, the self-polishing performance of the coatings was tested using dynamic acceleration experiments. The coating to be tested was applied to an ABS plate (25 cm × 10 cm × 0.3 cm) with a coating thickness of about 300 μm (Ni et al., 2020). The dried sample

Table 2
Basic formula of antifouling paints.

Component	Mass fraction(%)
Resin	40–50
Cu ₂ O	15–40
Copper pyridyl thione	0–15
Iron oxide red	4–8
ZnO	5–10
Talcum powder	4–8
Thixotropic agent	1–2
xylene	5–10

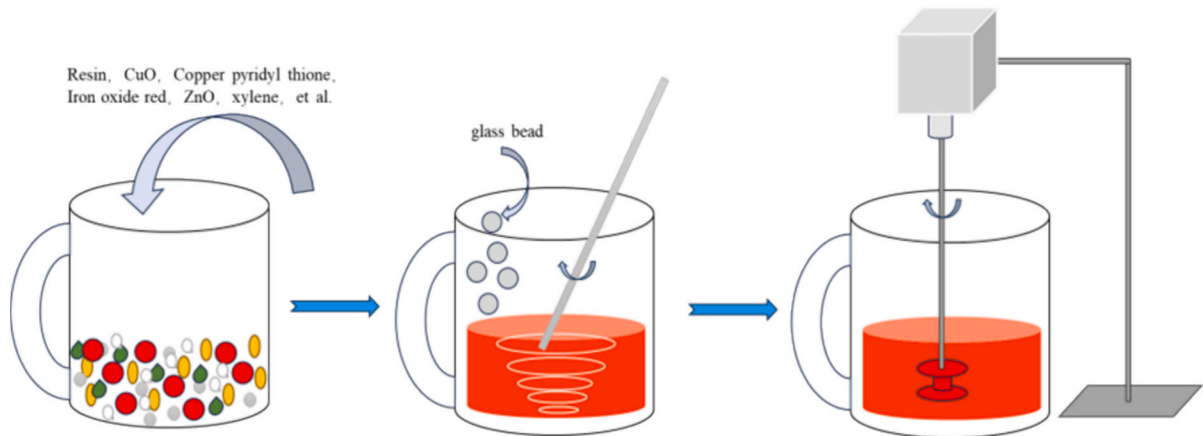


Fig. 2. Formulation of antifouling coatings.

Table 3
Prepared antifouling paints.

Antifouling paints	Cu ₂ O(%)	Copper pyridyl thione (%)
P-B/N-0	30	10
P-B-3	20	0
P-B-6	20	0
P-B-9	20	0
P-N-3	20	0
P-N-6	20	0
P-N-9	20	0

Table 4
Formula of nutrient solution.

Algae	Nutrient salts/volume fraction				
	Vitamins	Sodium nitrate	Sodium silicate	Trace element	Sodium phosphate
Content (μL/ 100 mL seawater)	50	100	100	50	100

plates were fixed in a laboratory dynamic accelerated experimental setup with seawater at a temperature of about 16 °C, a pH of 8.2, and a rotational speed of 130 rpm. 6 cycles of testing were performed, one cycle every 7 days, and at the end of each experimental cycle, the coatings were washed with deionized water, dried in an oven at 40 °C, and the surface morphology of each cycle coating was observed using a laser confocal microscope and its weight was weighed using an electronic balance.

2.7. Anti-protein adsorption tests of antifouling paints

Coomassie brilliant blue method is commonly used to determine the BSA adsorption of antifouling paints (Kielkopf et al., 2020; Qin et al., 2023). The standard working curve of BSA was first determined. Dilute BSA to 6 different concentrations with a certain concentration of sodium chloride solution, and then position the wavelength of UV-visible spectrophotometer to 595 nm, and measure the absorbance value of each concentration at this wavelength. The absorbance value-concentration of BSA was plotted to obtain the linear regression equation, which is the standard working curve of BSA. The coating was then placed in a surface dish with 40 mL of PBS buffer solution and ultrasonically cleaned for 30 min. The coatings were removed and wetted with 40 mL of PBS for 2 h. The wetted coatings were immersed in 10 mL of BSA (1 mg/mL) solution and incubated at 20 °C for 4 h. After

completion of the incubation, the coatings were removed and the coatings were carefully washed with 10 mL of PBS (0.1 M) to remove the proteins from the surface of the coatings. The washing solution was stained with Coomassie brilliant blue, and the absorbance value of the washing solution was measured at 595 nm. Finally, the BSA concentration in the solution was calculated using a standard curve, which in turn gave the amount of BSA adsorbed per unit area. The above experiment was repeated three times and the average value was taken (Zhou et al., 2021).

2.8. Antifouling test in marine

Antifouling test in a marine environment is necessary to evaluate the antifouling properties of antifouling paints (Qian et al., 2010). Antifouling paints were coated on the PVC panels (300 mm × 150 mm × 6 mm) with the thickness of 600 μm. The prepared sample plates were placed in a cool and ventilated place to dry naturally, and then immersed into 1 m deep seawater in Qingdao harbor, and fished out once every six months to check, and then carefully cleaned with seawater after fished out, and photographs were taken to record the number of fouling organisms.

3. Results and discussion

3.1. IR of acrylate resins containing indole derivative structure

The peak of 3387 cm⁻¹ was related to the stretching vibration absorption peaks of -NH-, the stretching vibration absorption peaks of -CH₃ and -CH₂ appeared at 2963 cm⁻¹ and 2867 cm⁻¹. The stretching vibration absorption peaks of carbonyl group in the carboxylic acid appeared at 1733 cm⁻¹, and the symmetric and asymmetric stretching vibration absorption peaks of -COO- appeared at 1023 cm⁻¹ and 1257 cm⁻¹ respectively. This shows that the indole structure has been introduced into the prepolymer and acrylate resin. In addition, In the IR spectrum of acrylate resin containing indole derivative structure, the stretching vibration absorption peaks of -COO-Zn-COO- appeared at 1601 cm⁻¹, and the stretching vibration absorption peak at 715 cm⁻¹ and 420 cm⁻¹ was according to Zn—O. However, there were no such absorption peaks appeared in the prepolymer, which indicated that acrylate resin with Indole derivative structure was successfully synthesized (Fig. 4).

3.2. Antibacterial activity evaluation of zinc acrylic resins containing indole derivative structure

The antibacterial activity test results are shown in Fig. 5. Compared

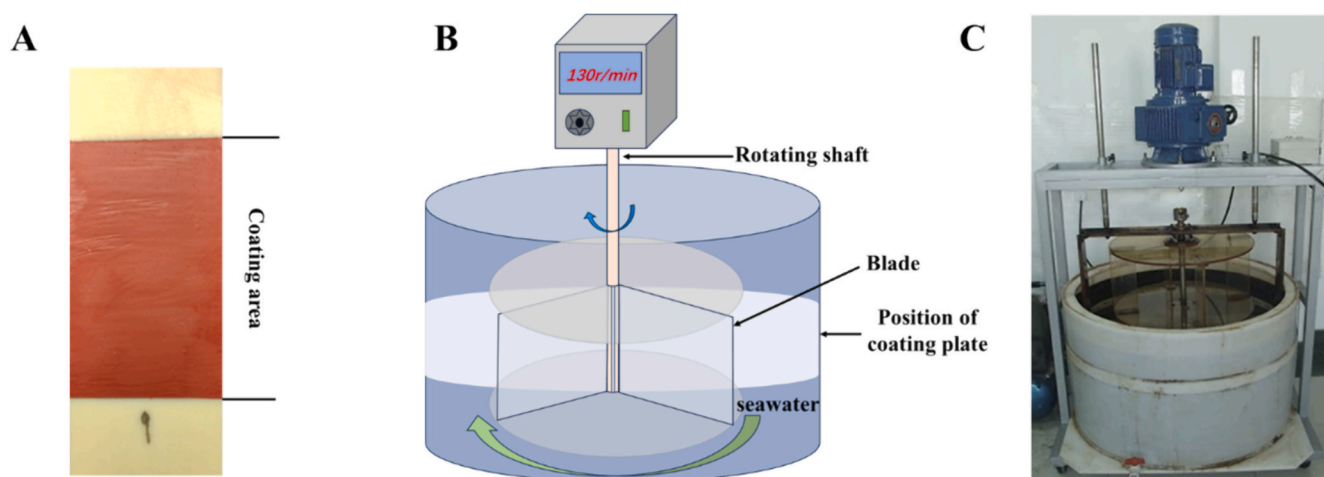


Fig. 3. Schematic of ABS panels (A) Dynamic simulation experiment device (B); Practical image of dynamic simulation experiment device (C).

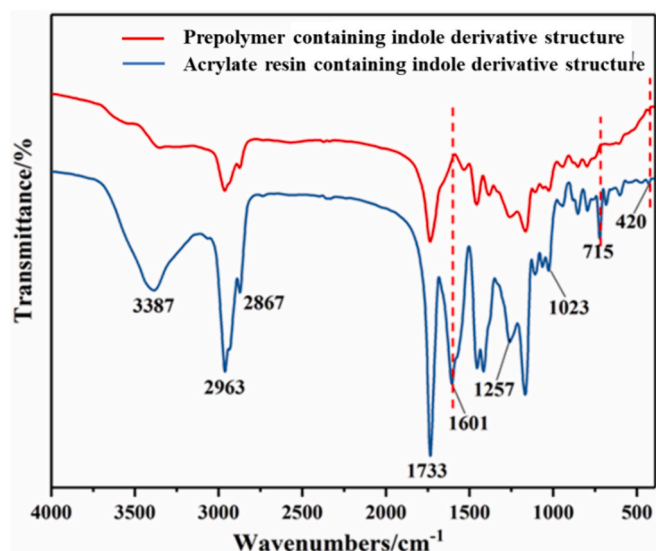


Fig. 4. IR of prepolymer and acrylate resin containing indole derivative structure.

with the resin without indole derivative structure, the antibacterial rate of the resin with indole derivative structure against *E. coli* and *P. tetraodonis* was significantly improved. The inhibition rates of R-B-9 against *E. coli* and *P. tetraodonis* were 98 % and 97 %, respectively, while the inhibition rates of R-B-0 against *E. coli* and *P. tetraodonis* were only 70 % and 68 %, respectively. At the same time, both the inhibition rates of R-N-9 against *E. coli* and *P. tetraodonis* had reached 98 %, but R-N-0 were only 71 % and 69 %, respectively.

3.3. Algae inhibition properties of zinc acrylate resins containing indole derivative structures

As shown in Fig. S1, 674 nm, 687 nm and 682 nm are the maximum absorption wavelengths of *Gymnodinium aerucyinosum*, *Phaeodactylum*

tricornutum Bohlin and *Prymnesium parvum*, respectively. Fig. S2 shows the standard working curves of the three microalgae, which proves that there is a good linear relationship between the absorbance at the maximum absorption wavelength and the concentration of the microalgal liquid, and therefore the absorbance can be used as a quantitative criterion for the concentration of the microalgal liquid. Fig. S3 shows microscope images of three algal cells. The results of algae inhibition are shown in Fig. 6.

As can be seen from Fig. 6, the inhibition of the three microalgae by the resin without indole derivative structure was in the range of 60–70 %, while the inhibition of the three microalgae by the two acrylic resins containing indole derivative structure was in the range of 80–90 %. The inhibition rate of the resins increased with the increase of BIAM content. When the BIAM content was 9 %, the growth inhibition rate of the three microalgae was the highest, and the inhibition rate of R-B/N-9 resin on *Prymnesium parvum* was up to 90 %.

3.4. Self-polishing properties of antifouling paints

The self-polishing properties of the resins were tested by dynamic simulation experiments using abrasion rate as evaluation criteria. The abrasion rates of the coatings containing Indole derivative structures are shown in Fig. 7. The highest abrasion rates of all the coatings were observed in the first cycle due to the release of small molecules remaining on the surface of the coatings as well as solvents. Starting from the second cycle, due to the uniform hydrolysis and polishing of zinc acrylate resin, the abrasion rate of the coating remains stable.

The laser confocal microscopy scanning images of the coatings containing Indole derivative structures are shown in Fig. 8. After five cycles of testing in the dynamic simulation experiment, the surface morphology of the coatings changed less, and none of them showed phenomena such as cracking and peeling. Therefore, the introduction of BIAM did not affect the good polishing performance of the coating.

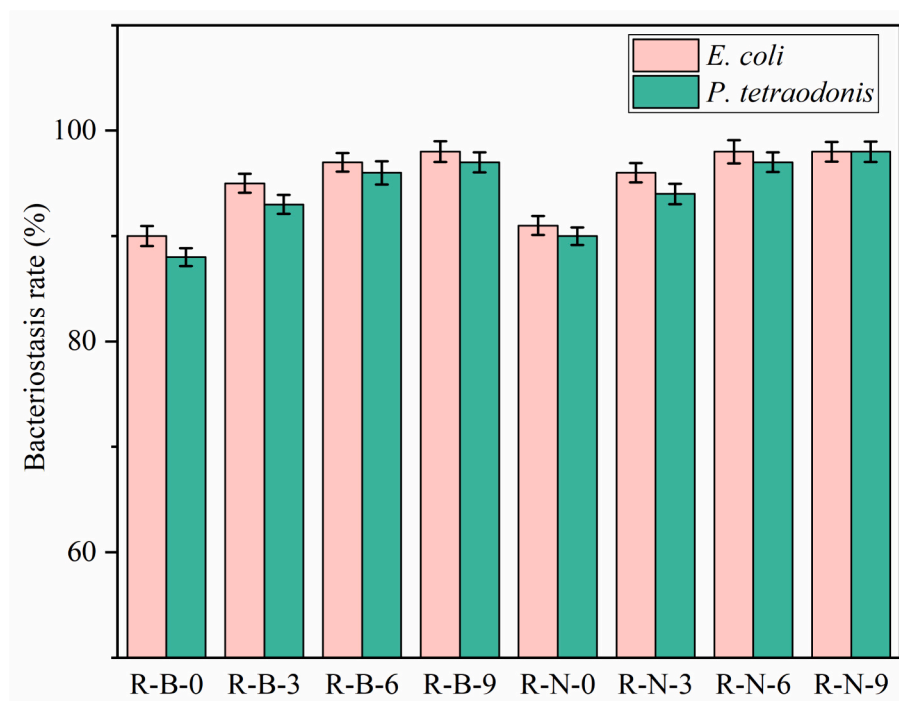


Fig. 5. Bacteriostasis rate (%) of resins against *E. coli* and *P. tetraodonis*.

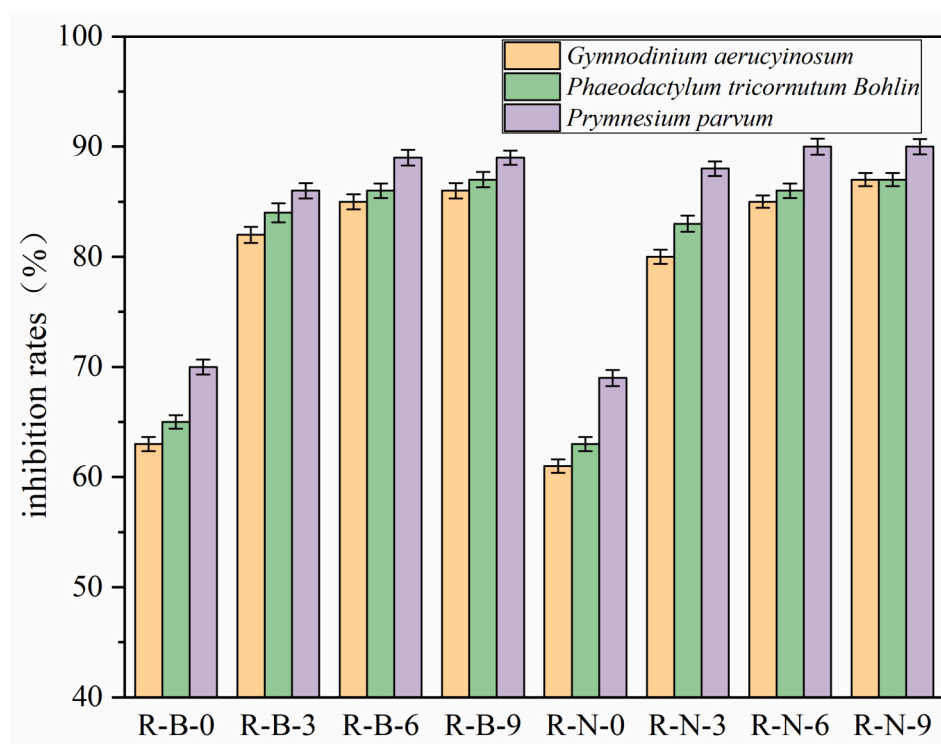


Fig. 6. Inhibition rats of zinc acrylic resins containing indole derivative structure.

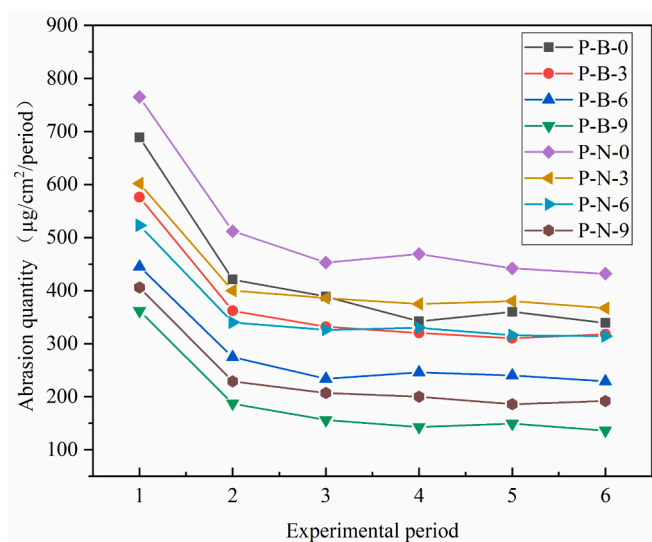


Fig. 7. The abrasion quantity of antifouling coatings.

3.5. Anti-protein adsorption properties of antifouling paints

The antifouling coatings were tested for their resistance to protein adsorption using UV spectrophotometry. As shown in Fig. 9, the standard working curve of BSA solution showed a linear relationship between protein concentration and optical density, and the linear curve fit was good.

In the process of biofouling formation, the first stage is mainly polysaccharides and proteins adsorbed on the surface of the material to form an initial biofilm, which increases the surface roughness of the material and is a prerequisite for the formation of the second stage of biofouling. Therefore, inhibiting the formation of protein membrane can effectively prevent the occurrence of fouling. Many studies have shown

that the anti-protein adsorption properties of materials are related to their surface energy, surface morphology or surface wettability (Kwon et al., 2020; Ma et al., 2015; Wang et al., 2020; Zhao et al., 2014). As can be seen in Fig. 10, the adsorption of BSA decreases significantly with the increase of BIAM content in the coating. This is because there are hydrophilic groups on the surface of the coating based on zinc acrylate resin, which has a certain inhibitory effect on protein adhesion. When BIAM is introduced, the hydrophobicity of the coating increases, and the hydrophobicity becomes stronger with the increase of BIAM content (Fig. S2.). This leads to an increasingly unstable coating surface, making it more difficult for proteins to adhere to the coating surface proteins (Ni et al., 2020). Therefore, the introduction of BIAM can significantly reduce the adsorption amount of BSA.

3.6. Marine antifouling performance of antifouling paints

The results of the antifouling coatings immersed in seawater for 6, 12, 18 and 24 months are shown in Fig. 11. In the blank group, the attachment of large fouling organisms had already appeared after 6 months of immersion in sea water, and the surface of PVC boards had been completely covered by algae and other fouling organisms after 24 months. In addition, in the control group, P-B-0 and P-N-0 adhered to a large amount of seaweed and sludge on the surface of the coating after 6 months of immersion. In the experimental group, except for P-B-3 and P-N-3, the other coatings had smooth surfaces with only a small amount of sludge attached. After 12 and 18 months of seawater immersion, the surfaces of coatings P-B-0 and P-N-0 have been covered by a large number of algae, the surfaces of coatings P-B-3, P-N-3, P-B-6, and P-N-6 have only a small amount of algae attached, and there are almost no fouling organisms attached to the surfaces of coatings P-B-9 and P-N-9. After 24 months of seawater immersion, the surfaces of coatings P-B-0 and P-N-0 begin to show large-scale fouling organisms attached, while the surface of coatings P-B-9, P-N-6, P-N-9 was clean with only a little algae or sludge attached. The long-term antifouling performance of these coatings is still under further testing.

The experimental results indicate that the introduction of BIAM into

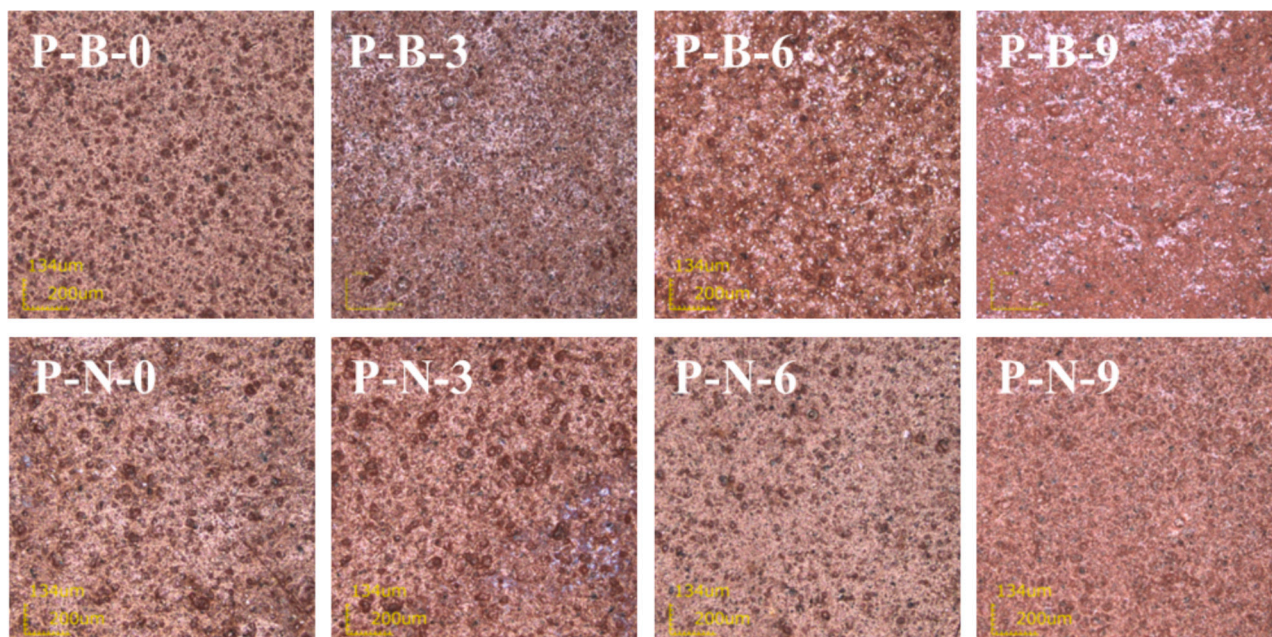


Fig. 8. Surface morphology of antifouling coatings after dynamic experiment.

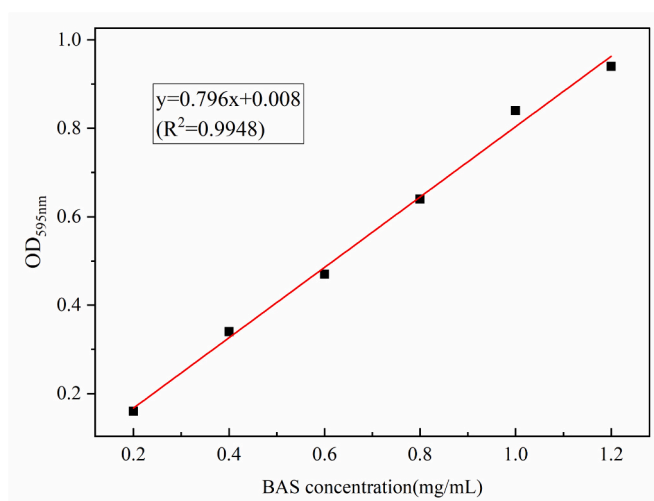


Fig. 9. Standard work curves of BSA solution.

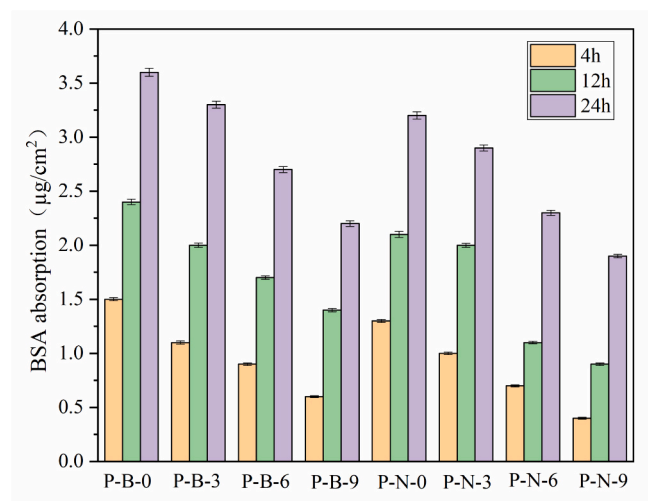


Fig. 10. BSA adsorption on marine antifouling paints.

acrylic resin enhances the antifouling performance of the coating, and the higher the proportion of indole derivative structures in the coating, the better the antifouling performance of the coating.

3.7. Study on antifouling mechanism

The antifouling mechanism of self-polishing antifouling coatings containing Indole derivative structures is shown in Fig. 12. The coating has a highly biologically active Indole derivative structure that makes it difficult for fouling organisms to adhere to the surface of the coating. Studies have shown that calcium ions are involved in almost all physiological activities of algae and are closely related to the growth and development of microorganisms. And indole derivatives can interfere with the balance of calcium ions (Ca^{2+}) in algal cells and reduce the concentration of Ca^{2+} in the cells, thus inhibiting the growth of algae (Feng et al., 2019c). In addition, the metal carboxylate group ($-\text{COO}-\text{Zn}-\text{COO}-$) in the coating undergoes a hydrolysis reaction with Na^+ , Cl^- , etc. in seawater, generating hydrophilic carboxyl groups on the surface,

making the coating surface hydrophilic. Then the coating surface peels off under the action of seawater erosion, exposing new coatings, thus achieving continuous renewal of the coating surface. Therefore, self-polishing antifouling coatings containing Indole derivative structures combine the high inhibitory bioadhesion activity of the compounds, the self-polishing property of the acrylic resin, and can obtain better anti-fouling performance while reducing the amount of Cu_2O .

4. Conclusions

Zinc acrylate resins with side-chain-suspended Indole derivative structures and self-polishing antifouling coatings based on them were prepared. After the introduction of the indole functional structure, the antibacterial and algal inhibition properties of the resin were significantly improved. When the content of BIAM was 9 %, the inhibition rates of *E. coli* and *Prymnesium parvum* could reach 98 % and 90 %, respectively. After the introduction of indole functional structure, the anti-protein adsorption performance of the coating was improved. With

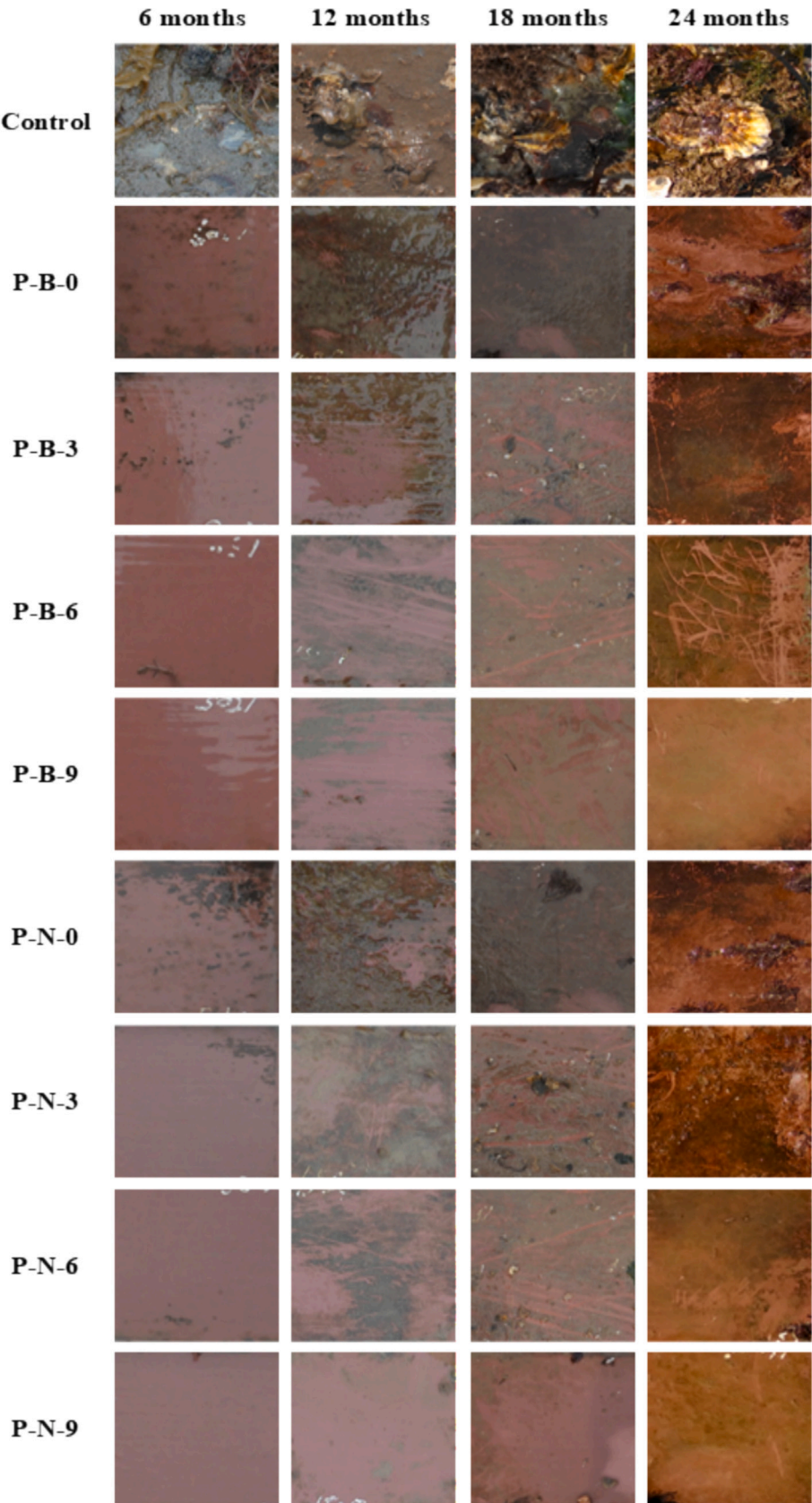


Fig. 11. Images of the coating after immersion in seawater for 6, 12, 18, and 24 months.

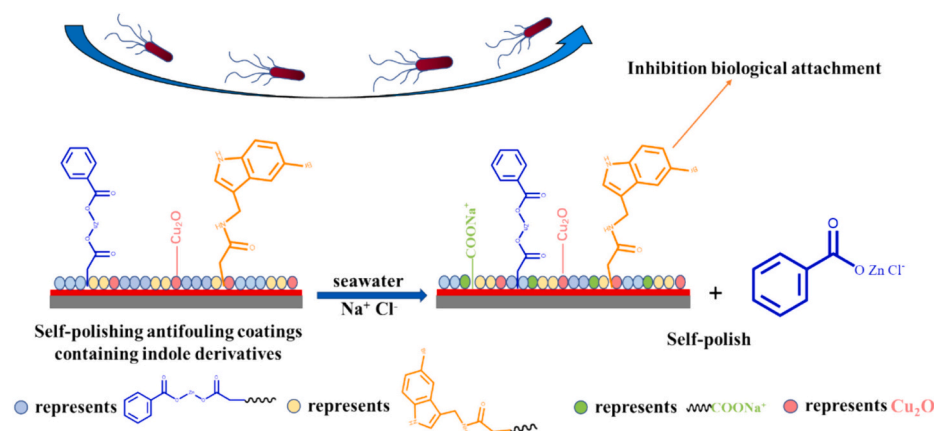


Fig. 12. Antifouling mechanism of self-polishing antifouling coatings containing indole derivatives.

the increase of BIAM content, the antifouling performance of the antifouling coating was better. After two years of offshore testing, the surface of the coating remained nearly smooth.

In conclusion, the new self-polishing antifouling coatings prepared in this study can significantly reduce the use of cuprous oxide in the current self-polishing antifouling coatings, and its long-lasting antifouling performance is also worthy of expectation.

CRedit authorship contribution statement

Wenjian Dong: Writing – original draft, Validation. **Chunhua Ni:** Methodology, Investigation. **Xia Li:** Funding acquisition, Data curation, Conceptualization. **Liangmin Yu:** Supervision, Resources. **Xuefeng Yan:** Visualization, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.marpolbul.2024.116931>.

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